Airbnb Analytics: Pricing & Performance Insights

ASIAN SCHOOL OF MEDIA STUDIES
SCHOOL OF DATA SCIENCE

Diploma
in
Data Science

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Under the Supervision of

ASIAN SCHOOL OF MEDIA STUDIES
NOIDA

2025

Abstract

This project, titled "Airbnb Analytics: Pricing and Performance Insights," aims to analyze Airbnb listings data to uncover key trends and insights in the short-term rental market. Utilizing Python for data cleaning and exploratory data analysis (EDA), we will examine factors influencing pricing, occupancy rates, and customer preferences across various neighborhoods.

By leveraging datasets from Inside Airbnb, we will identify patterns related to pricing, amenities, and seasonal variations. Interactive visualizations will be created using Power BI and Excel to present findings effectively. Additionally, predictive modeling will forecast rental prices and occupancy rates, providing actionable insights for hosts to optimize their listings.

The ultimate goal is to equip Airbnb hosts and guests with data-driven recommendations, enhancing decision-making and improving the overall experience in the competitive short-term rental landscape.

Motivation

The rise of the sharing economy has revolutionized the way people travel and experience hospitality, with Airbnb at the forefront of this transformation. As the platform continues to grow, understanding the dynamics of pricing and performance becomes crucial for both hosts and guests. With millions of listings worldwide, the ability to analyze and interpret data can provide significant advantages in a competitive market.

This project is motivated by the need to empower Airbnb hosts with actionable insights that can enhance their rental strategies, optimize pricing, and improve guest satisfaction. By leveraging data analytics, we can uncover hidden trends and correlations that inform better decision-making. Additionally, potential guests can benefit from a deeper understanding of pricing patterns and availability, leading to more informed choices when booking accommodations.

In an era where data-driven decisions are paramount, this analysis aims to bridge the gap between raw data and practical insights, ultimately contributing to a more efficient and enjoyable Airbnb experience for all stakeholders involved.

Literature Review

Research on Airbnb has highlighted several key factors influencing short-term rental markets. Reviews and ratings are also crucial, as higher ratings correlate with increased occupancy and pricing power.

Seasonal trends further influence occupancy rates, with demand peaking during tourist seasons. Additionally, the use of data analytics, including machine learning for price prediction, is increasingly emphasized as a means for hosts to make informed decisions.

This project will leverage these insights by analyzing Airbnb listings data to provide actionable recommendations for hosts and guests, enhancing understanding of pricing and performance in the market

Methodology

1. Data Collection:
The project will utilize datasets from Inside Airbnb, encompassing listings, reviews, and availability information to provide a comprehensive view of the market.

2. Data Cleaning:
A rigorous cleaning process will be implemented to address missing values, outliers, and inconsistencies, ensuring the data's accuracy and reliability for analysis**.**

3. Exploratory Data Analysis (EDA):
Using Python libraries like Pandas and Matplotlib, EDA will visualize key variables and identify trends and relationships between pricing, occupancy rates, and listing attributes.

4. Predictive Modeling:
Machine learning algorithms, such as linear regression and decision trees, will be employed to forecast rental prices and occupancy rates, aiding hosts in optimizing their pricing strategies.

5. Visualization and Reporting:
Findings will be presented through interactive visualizations in Power BI and Excel, providing actionable insights and recommendations for Airbnb hosts to enhance their performance.

Planning of Work

Phase	Days	Work Done
Phase 1	1–2	Research and data collection
Phase 2	3–4	Cleaning and preparing the data
Phase 3	5–6	Building and testing the model
Phase 4	7	Improving and analyzing results
Phase 5	8	Final documentation and report

Facilities Required

Hardware:
- Laptop or computer with at least 8GB RAM and i5 processor.

Software and Libraries:
- Python (3.x)
- Jupyter Notebook
- Libraries: pandas, numpy, sklearn, nltk, matplotlib
- Power Bi
- Microsoft Excel

Datasets:
- Sample airbnb data from Kaggle

Project Outcomes

1. **Key Pricing Factors Identified**: Insights into how variables like location, room type, reviews, and availability impact Airbnb listing prices.
2. **Performance Metrics Analysis**: Evaluation of occupancy rates, average daily rates (ADR), and revenue patterns across different property types and regions.
3. **Predictive & Pricing Models:** Development of models to forecast optimal pricing and predict booking performance based on listing features.
4. **Visual Dashboards & Recommendations:** Interactive visualizations and actionable pricing strategies to support Airbnb hosts in maximizing revenue.

Bibliography

[1] **Airbnb Analysis EDA - Kaggle**

[2] **Syed-Adnan-S/Airbnb-Bookings-Analysis-Capstone-Project - GitHub**

[3] **Portfolio project || EDA on Airbnb NYC 2019 - YouTube**

[4] **Exploratory Data Analysis - Airbnb Bookings Dataset - LinkedIn**

[5] **Exploratory Data Analysis on NYC Airbnb 2019 dataset - Medium**

[6] **End-to-End Predictive Analysis on AirBnB Listings Data - Analytics Vidhya**

[7] **Airbnb EDA Project - GitHub**

[8] **Data Analysis Project: Scenario 1 - Chicago Airbnb 2019**

[9] **NYC Airbnb: EDA, Visualization, Regression - Kaggle**

[10] **My data analysis project with Purwadhika - LinkedIn**